

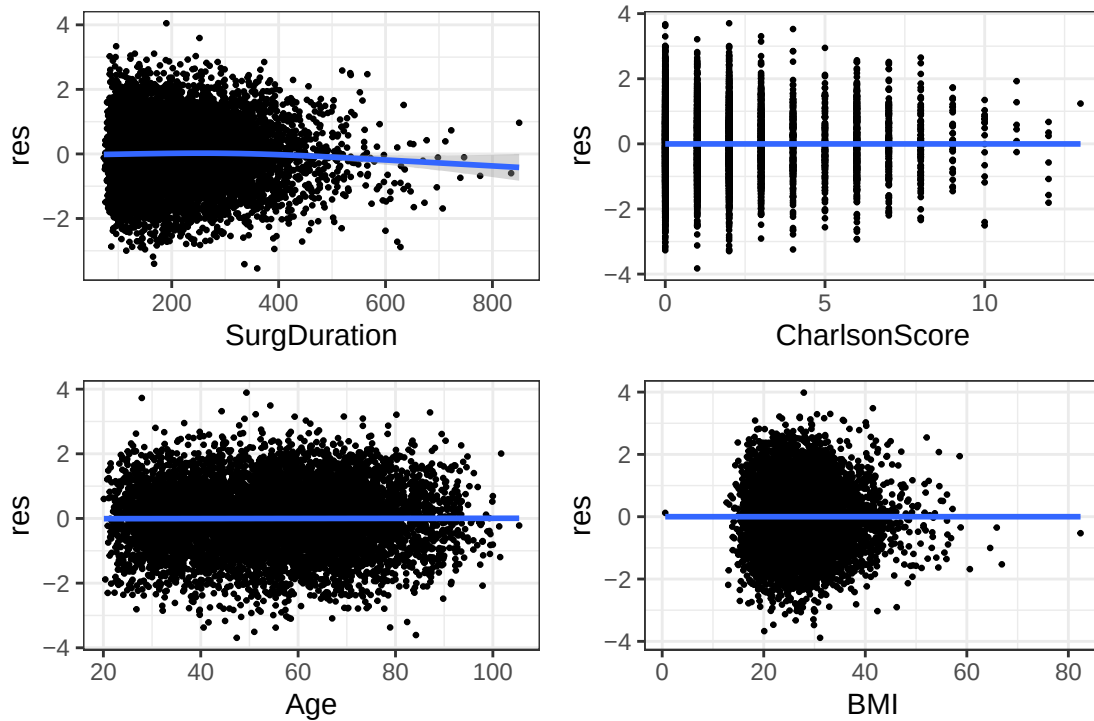
Model 3

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```
m3 <- glm(SeriousInfection ~ SurgDuration + Open +  
  Age + FEMALE + BMI + CharlsonScore + CHF + VALVE +  
  RENLFAIL + LIVER + METS + TUMOR + COAG +  
  OBESE + LYLES + BLDLOSS + ANEMDEF + DRUG +  
  SteroidHx + ImmunosuppressantHx,  
  data = ct, family = binomial)
```

Diagnostics

Quantile residual plots:



Cook's distance:

```
summary(cooks.distance(m3))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
8.081e-07	2.640e-06	6.125e-06	1.285e-04	1.892e-05	7.774e-03

Variance inflation factors:

```
car::vif(m3)
```

SurgDuration	Open	Age	FEMALE
1.054708	1.042331	1.361434	1.034689
BMI	CharlsonScore	CHF	VALVE
1.651082	7.179817	1.452949	1.286382
RENLFAIL	LIVER	METS	TUMOR
1.784761	1.152899	4.084421	2.049094
COAG	OBESE	LYTES	BLDLOSS
1.172340	1.618556	1.164501	1.024790
ANEMDEF	DRUG	SteroidHx	ImmunosuppressantHx
1.110125	1.043610	1.085643	1.041719

Model summary

```
summary(m3)
```

Call:

```
glm(formula = SeriousInfection ~ SurgDuration + Open + Age +  
    FEMALE + BMI + CharlsonScore + CHF + VALVE + RENLFAIL + LIVER +  
    METS + TUMOR + COAG + OBESE + LYTES + BLDLOSS + ANEMDEF +  
    DRUG + SteroidHx + ImmunosuppressantHx, family = binomial,  
    data = ct)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.2684731	0.2911630	-11.226	< 2e-16 ***
SurgDuration	0.0030954	0.0003612	8.570	< 2e-16 ***
Open	0.1889010	0.1183949	1.596	0.110597

Age	-0.0070918	0.0028394	-2.498	0.012502	*
FEMALE	-0.0979180	0.0833971	-1.174	0.240348	
BMI	0.0035796	0.0084405	0.424	0.671497	
CharlsonScore	0.0659385	0.0500038	1.319	0.187279	
CHF	0.2343756	0.2222662	1.054	0.291663	
VALVE	-0.1775948	0.1868224	-0.951	0.341803	
RENLFAIL	0.2206456	0.2162830	1.020	0.307648	
LIVER	-0.4447646	0.2740971	-1.623	0.104664	
METS	-0.5960454	0.3433462	-1.736	0.082566	.
TUMOR	-0.2179143	0.1531355	-1.423	0.154731	
COAG	0.4802779	0.1316000	3.650	0.000263	***
OBESE	-0.2304488	0.1409222	-1.635	0.101988	
LYTES	0.6085770	0.0889743	6.840	7.92e-12	***
BLDLOSS	0.1981421	0.2349280	0.843	0.398996	
ANEMDEF	0.0398281	0.0966510	0.412	0.680280	
DRUG	0.5363208	0.2239584	2.395	0.016632	*
SteroidHx	0.0727722	0.0886435	0.821	0.411672	
ImmunosuppressantHx	-0.0856989	0.2311782	-0.371	0.710858	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4608.2 on 7907 degrees of freedom
 Residual deviance: 4395.3 on 7887 degrees of freedom
 AIC: 4437.3

Number of Fisher Scoring iterations: 5